

Questions related to TC

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Program 1

```
public static void function(int n)
{
    int i = 1, s = 1;
    while (s < n)
    {
        s = s + i;
        i++;
    }
}
```

Loop will stop $s = n$

i	s
1	1
2	1+1
3	2+2
4	2+3
5	2+4
6	2+5
⋮	⋮
k	$\{s = n\}$

$$s_n = \frac{n(n+1)}{2}$$

$$s = 2+2+3+4+5+\dots+k-1$$

$$= 1 + (1+2+3+4+\dots+k-1)$$

$$= 1 + \frac{(k-1)(k-1+1)}{2}$$

$$s = 1 + \frac{k(k-1)}{2}$$

$$\Rightarrow n = 1 + \frac{k^2 - k}{2}$$

$$2n = 2 + k^2 - k$$

$$\Rightarrow k^2 - k = 2n - 2$$

$$\Rightarrow k =$$

Program 2

```
public static void fun(int n)
{
    if (n < 5)
        Console.WriteLine("Timecomplexity");
    else {
        for (int i = 0; i < n; i++) {
            Console.WriteLine(i + " ");
        }
    }
}
```

$O(n)$

Program 3

```
void fun(int n)
{
    for(int i=0; i*i<n; i++)
        cout<<"Timecomplexity";
}
```

$i \times i = n$ (stop)

$k \times k = n \Rightarrow k^2 = n$
 $k = \sqrt{n}$

TC = $O(\sqrt{n})$

Program 4

```
void fun(int n, int x)
{
    for (int i = 1; i < n; i = i * x)
        cout << "Timecomplexity";
}
```

$i < n$
 $x = 1/2$

$i > n$
 $i = n$

$O(\sqrt{x})$

No. of Iteration

1
2
3
...

i
 $1 \times x = x$
 $1 \times x \times x = x^2$
 $1 \times x \times x \times x = x^3$
 $\vdots$
 x^k

n

Program 5

```
for (i = 1; i <= n; i=i*2)
{
    Console.WriteLine("Hello World !!!\n");
}

while(i<=n)
{
    i=i*2;
}
```

iteration

1
2
3
4
...

i

1
1x2
1x2x2
1x2x2x2
2^{k-1}

$$n = 2^{k-1}$$

$$\log_2 n = \log_2 2^{(k-1)} \Rightarrow k-1 = \log_2 n \Rightarrow T$$

Program 6

```
void fun(int n)
{
    int i = 1;
    while (i < n) {
        i = i * 2;
    }
}
```

$$i = n$$

$$i = i * 2$$

$$i/2, i/2 \rightarrow O(\log_2 n)$$

Program 7

```
void fun(int j)
{
    while (j > 0) {
        j = j / 2;
    }
}
```

$$j = 0$$

1

No. of iteration

1
2
3

j

j/2^{k-1}
j/2^{k-2}
j/2^{k-3}

Program 8

```
void fun(int n)
{
    int i = 1;
    while (i < n) {
        int j = n;
        while (j > 0) {
            j = j / 2;
        }
        i = i * 2;
    }
}
```

$$\log_2 n$$

$$\rightarrow O(\log_2 n \times \log_2 n)$$

$$\begin{matrix} 4 \\ \vdots \\ k \end{matrix}$$

$$\begin{matrix} j/2^3 \\ j/2^{k-1} \end{matrix}$$